# **Design Challenge 2:** A Shell Company

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| **Preview:** | **Wed Jan 28** |
| **Assigned:** | **Fri Jan 30** |
| **Due:** | **Machine Demo: Fri Feb 13** (in Lab sections)<br>**Documentation: Wed Feb 18** (5:00 PM) |
| **Grade Weight:** | **15%** |

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# DC2: Operation Goal and Definitions

**Whole Sunflower Seeds** → **Shells** + **Kernels**

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# DC2: Objectives and Deliverables

## Design Objective

EGR Corp is looking for design teams to build and demonstrate a simple and effective “**Sunflower Shelling Machine”** (or SSM), a mechanical device that, through manual means only, shells and separates sunflower seeds.

## Two Deliverables:

1. **Design, Build, Test, and Demonstrate SSM performance** **= 70% (**50% + 10% + 10%) of total Score

2. **Written Documentation in Memo Format** **= 30%** of total Score

**2.1 Executive Summary**

**2.2 Approach Description**

Ideation techniques used, and why:

- Brainstorming technique and/or MindMapping

Idea selection and refinement techniques used, and why:

- Voting, Value Matrix and why
- Morphological Matrix
- How-Why Diagram

**2.3 3D Machine Sketch -** includes labeling all major components, sub-systems, and key features

**2.4 Appendix** – including copies of design process materials (e.g., lists, notes, and sketches)

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# DC2: Scoring Method

## I. Machine Performance Score (70%)

**(50%) *Performance Score* = Successful Shelling Rate = $\dot{S}$, over the 3-min Performance Window**

Where:

$$\dot{S} = \Big[\big(\text{Yield}\big) - \big(\text{Waste} \times 0.5\big)\Big] \Big/ \text{3 Minutes} = \text{Seeds/minute}$$

***Yield = # of Clean Kernels***

<u>"Clean Kernel" Definition</u>
- Kernel removed completely from Shell
- No damage to Kernel
- No material adhering to seed (no coatings, liquids, or solids)
- Minor damage (e.g., scuffs) to kernel coat is acceptable

***Waste = # of Unprocessed Seeds***

<u>"Unprocessed" Definition</u>
- Damaged kernel
- Partial / Incomplete Shelling
- No shelling

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# DC2: Scoring Methods

## I. Machine Performance Score (70%), Continued

**(10%) *Aesthetics and Craftsmanship***

Overall build quality, and attention to design and build details

**(10%) *Innovation***

Surprising and novel aspects and other technical and process risk-taking strategies. These could be:

**“Hardware”** (e.g., machine functional automation, system integration, and appropriate material choices)

**“Software”** (e.g., set up and performance process strategies, team coordination, and fault/failure management)

## II. Written Documentation Score (30%)

1. *Completeness*
2. *Clarity*
3. *Conciseness*

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# DC2: Rules

1. **Demo(1):** teams will be allotted a **7-minute** overall time slot to set up and demonstrate SSM performance. Within this time slot, teams have up to two three-minute performance windows, with the highest score achieved from both performance windows being retained.

2. **Demo(2):** teams are allowed multiple attempts, restarts, and machine adjustments (including device clearing/cleaning) within each performance window. No direct shelling operation help is allowed though.

3. **Demo(3):** SSM must be solely human-powered and operate while situated on a Lab tabletop (i.e., no stored or other energy methods allowed). In addition, seeds may not be touched/handled during performance window.

4. **Demo(4):** All Whole seeds to be shelled must be delivered from a single hopper within the SSM. Whole seeds may only be added to the hopper before and/or after a performance window.

5. **Demo(5):** Seeds should be processed in a manner that facilitates easy seed counting and scoring.

6. **Size(1):** The assembled SSM must fit entirely within a standard Foundry storage bin (with lid on).

7. **Size(2):** SSM may use deployable (e.g., “fold out”, but non-separable) elements that help enable machine operation and for assistance in meeting the **Size(1)** constraint above.

8. **Matl’s(1):** Wood, PVC, and cardboard (+ others) will be supplied as primary construction materials, however, teams may also use any free/found Foundry materials. You may also buy other materials as long as your total team “out of pocket” expenditure does not exceed $15 total.

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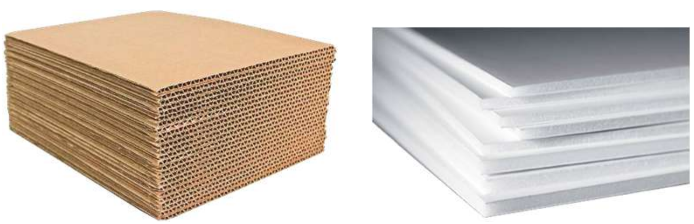
# DC2: Tools and Typical Materials

**X-Acto Knife**
(and blades)

**Hotmelt Glue Gun**
(and glue sticks)

**Assorted (typical) Materials Supplied**

- Wood dowels
- Elastic bands
- Wood sheet
- Wood lumber
- String
- Sandpaper (in various grades)
- PVC pipe (in various diameters)



**Cardboard / Foamcore Sheets**

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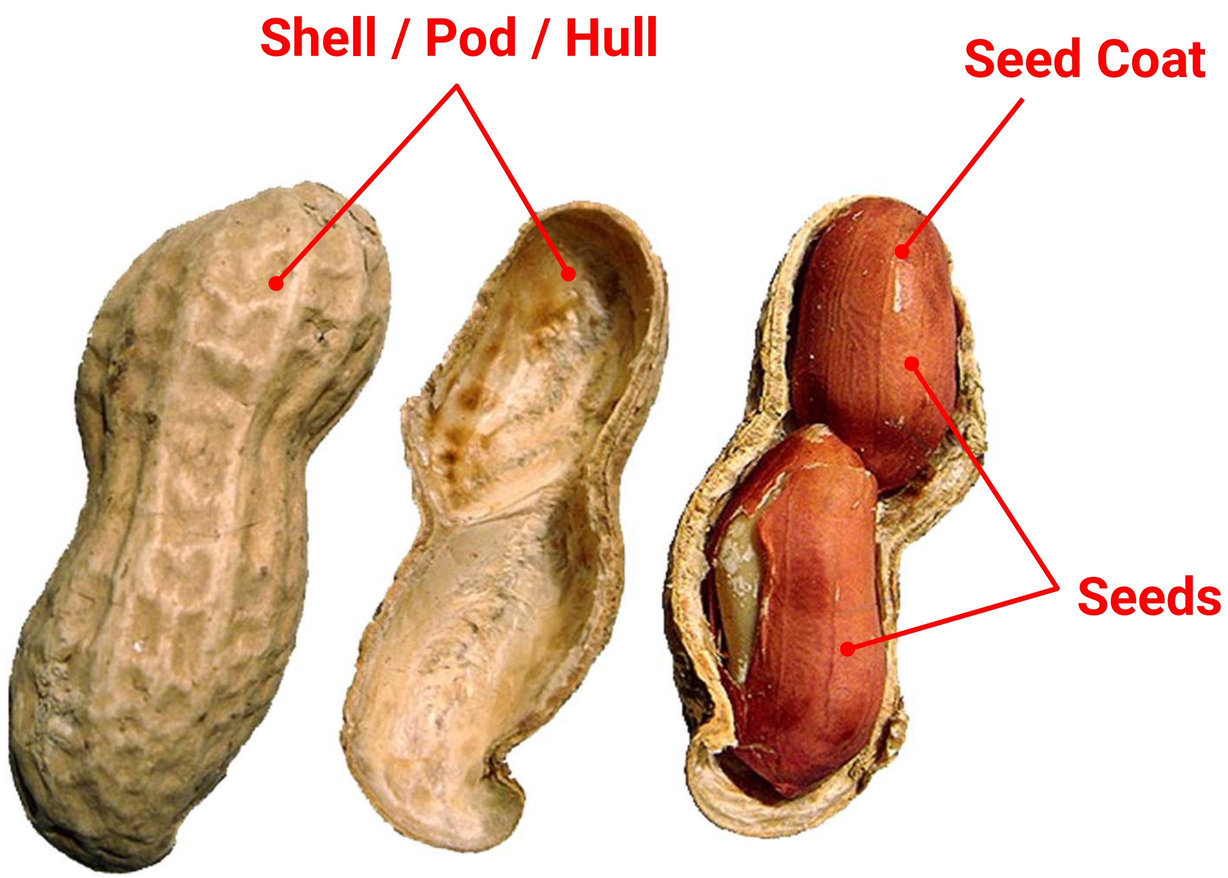
# DC2: Sunflower Terms

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